

Estimating Treatment Effects - Regression

March 10, 2026

Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$

Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$
- $ATE_{est} = Avg_n[Y^1|D = 1] - Avg_n[Y^0|D = 0]$

Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$
- $ATE_{est} = Avg_n[Y^1|D = 1] - Avg_n[Y^0|D = 0]$
- When $(Y^1, Y^0) \not\perp D$:

Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$
- $ATE_{est} = Avg_n[Y^1|D = 1] - Avg_n[Y^0|D = 0]$
- When $(Y^1, Y^0) \not\perp D$:

$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y^0|D = 1] - Avg_n[Y^0|D = 0]\}}_{\text{Selection Bias}} \\ + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias}}$$

Estimating Treatment Effects Review

- $ATE = Avg_n[Y^1 - Y^0]$
- $ATE_{est} = Avg_n[Y^1|D = 1] - Avg_n[Y^0|D = 0]$
- When $(Y^1, Y^0) \not\perp D$:

$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y^0|D = 1] - Avg_n[Y^0|D = 0]\}}_{\text{Selection Bias}} \\ + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias}}$$

- Solution?

- Randomize treatment exposure so that $(Y^1, Y^0) \perp\!\!\!\perp D$.

$$\begin{aligned} ATE_{est} = ATE + & \underbrace{\{Avg_n[Y^0|D=1] - Avg_n[Y^0|D=0]\}}_{\text{Selection Bias} = 0} \\ + & \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias} = 0} \end{aligned}$$

Estimating Treatment Effects Review

- Randomize treatment exposure so that $(Y^1, Y^0) \perp\!\!\!\perp D$.

$$\begin{aligned}ATE_{est} = ATE + & \underbrace{\{Avg_n[Y^0|D=1] - Avg_n[Y^0|D=0]\}}_{\text{Selection Bias} = 0} \\ & + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias} = 0}\end{aligned}$$

- Added bonus when LLNs applies:

Estimating Treatment Effects Review

- Randomize treatment exposure so that $(Y^1, Y^0) \perp\!\!\!\perp D$.

$$\begin{aligned}ATE_{est} = ATE + & \underbrace{\{Avg_n[Y^0|D=1] - Avg_n[Y^0|D=0]\}}_{\text{Selection Bias} = 0} \\ & + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias} = 0}\end{aligned}$$

- Added bonus when LLNs applies: Sample ATE = Population ATE

$$\{Avg_n[Y^1|D=1] - Avg_n[Y^0|D=0]\} = \{E[Y^1|D=1] - E[Y^0|D=0]\}$$

Estimating Treatment Effects Review

- So randomization is the ideal and the “gold standard” of research methodologies.

Estimating Treatment Effects Review

- So randomization is the ideal and the “gold standard” of research methodologies.
- But...in policy analysis it's often (though not always) impossible to randomize individuals in our sample into treatment?

Estimating Treatment Effects Review

- So randomization is the ideal and the “gold standard” of research methodologies.
- But...in policy analysis it's often (though not always) impossible to randomize individuals in our sample into treatment?
- How can we attempt to identify causal effects of policy when randomization is infeasible?

Estimating Treatment Effects - Selection into Treatment



Estimating Treatment Effects - Selection into Treatment

Comparison of Key Characteristics by Insurance Status					
Insurance Status	Mean Age	% Female	% Smoker	% Binge Drinker	% Earning Below \$50K
Uninsured	40.3	42.8%	21.1%	20.3%	52.2%
Medicaid	42.0	65.5%	24.7%	13.7%	67.1%

- If Medicaid were randomly assigned and our sample size was sufficiently large, what would we expect to see in this comparison of key characteristics by insurance status?

Estimating Treatment Effects - Selection into Treatment

Comparison of Key Characteristics by Insurance Status					
Insurance Status	Mean Age	% Female	% Smoker	% Binge Drinker	% Earning Below \$50K
Uninsured	40.3	42.8%	21.1%	20.3%	52.2%
Medicaid	42.0	65.5%	24.7%	13.7%	67.1%

- If Medicaid were randomly assigned and our sample size was sufficiently large, what would we expect to see in this comparison of key characteristics by insurance status?
- If we were able to calculate the true ATE of Medicaid coverage on health, do you expect that it would differ from the ATT?

Estimating Treatment Effects - Regression Analysis

Estimating Treatment Effects - Regression Analysis

- Assume that when key observed variables have been “made equal” across treatment and control groups...

Estimating Treatment Effects - Regression Analysis

- Assume that when key observed variables have been “made equal” across treatment and control groups...
- ...omitted variable bias from unobserved confounders will also be eliminated.

Estimating Treatment Effects - Regression Analysis

- Assume that when key observed variables have been “made equal” across treatment and control groups...
- ...omitted variable bias from unobserved confounders will also be eliminated.
- $ATE_{est} = Avg_n[Y^1|D = 1, X] - Avg_n[Y^0|D = 0, X]$

Estimating Treatment Effects - Regression Analysis

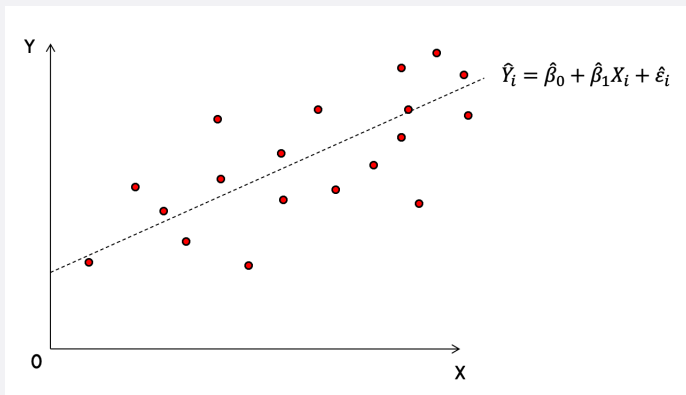
- Assume that when key observed variables have been “made equal” across treatment and control groups...
- ...omitted variable bias from unobserved confounders will also be eliminated.
- $ATE_{est} = Avg_n[Y^1|D = 1, X] - Avg_n[Y^0|D = 0, X]$
- Even if D is not randomly assigned, we can still *hope* that $(Y^1, Y^0) \perp\!\!\!\perp D$ conditional on the X 's.
 - ▶ (And that $ATE_{est} = ATE$)

Estimating Treatment Effects - Regression Analysis

- Assume that when key observed variables have been “made equal” across treatment and control groups...
- ...omitted variable bias from unobserved confounders will also be eliminated.
- $ATE_{est} = Avg_n[Y^1|D = 1, X] - Avg_n[Y^0|D = 0, X]$
- Even if D is not randomly assigned, we can still *hope* that $(Y^1, Y^0) \perp\!\!\!\perp D$ conditional on the X 's.
 - ▶ (And that $ATE_{est} = ATE$)
- BUT THIS NOT “GUARANTEED” LIKE IT IS WITH RANDOMIZATION AND DEPENDS ON A STRONG (UNTESTABLE) ASSUMPTION.

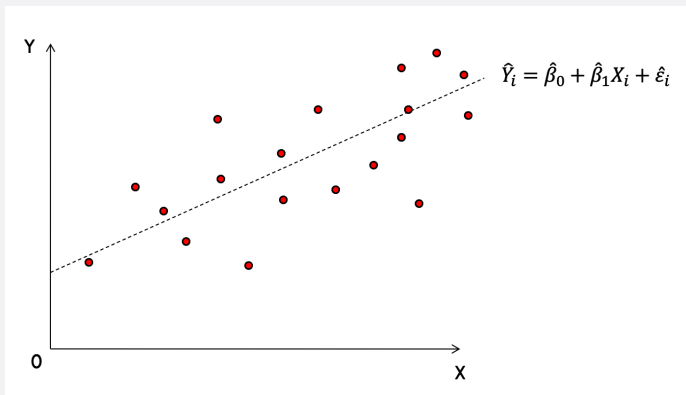
Estimating Treatment Effects - Regression Analysis

- So how does regression actually work?



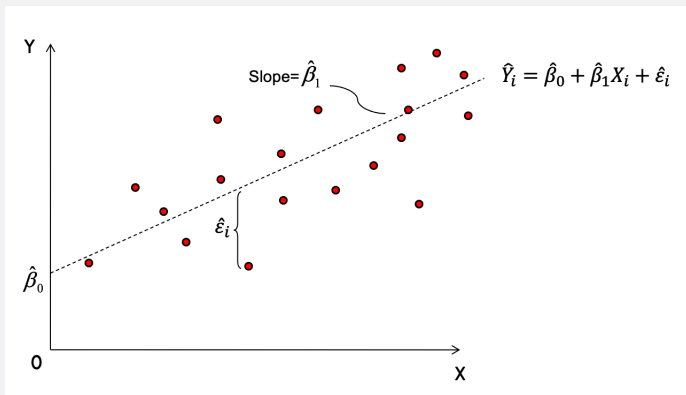
Estimating Treatment Effects - Regression Analysis

- So how does regression actually work?

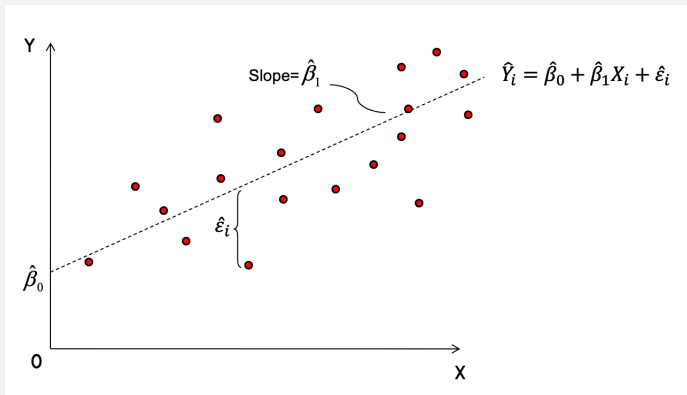


- Note that we have not randomized people into X in this case. We simply observe an individual's value of X in our sample.

Estimating Treatment Effects - Regression Analysis



Estimating Treatment Effects - Regression Analysis



- Most popular method for drawing the regression line involves minimizing the squared sum of the residuals. (Why the squared sum and not just the sum?)

Estimating Treatment Effects - Regression Analysis

- Suppose that we estimate a linear regression model of the following form: $y_{it} = \beta_0 + \beta_1 Edu_{it} + \varepsilon_{it}$

Where y is the number of times that individual i visits the emergency department in year t

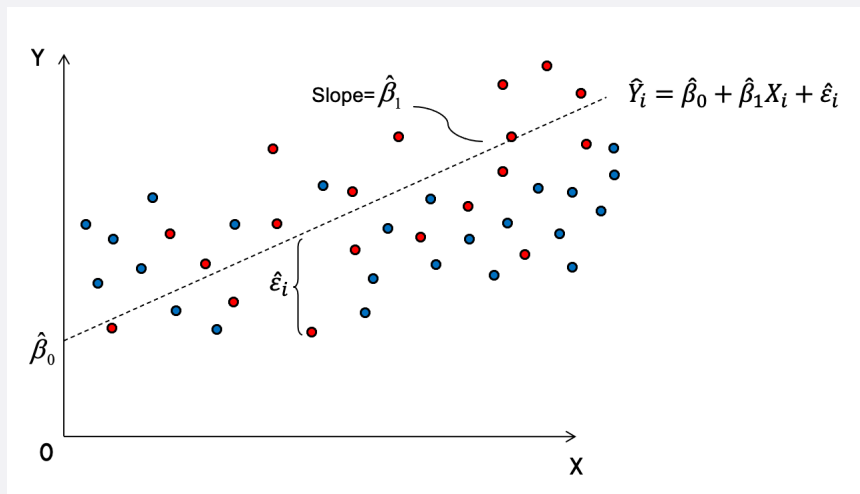
Edu is individual i 's education in years (e.g. a value of 12 represents a high school education, a value of 14 represents a college education, etc.).

Our coefficient estimates are as follows: $\hat{\beta}_0=4.0$ and $\hat{\beta}_1=-0.25$.

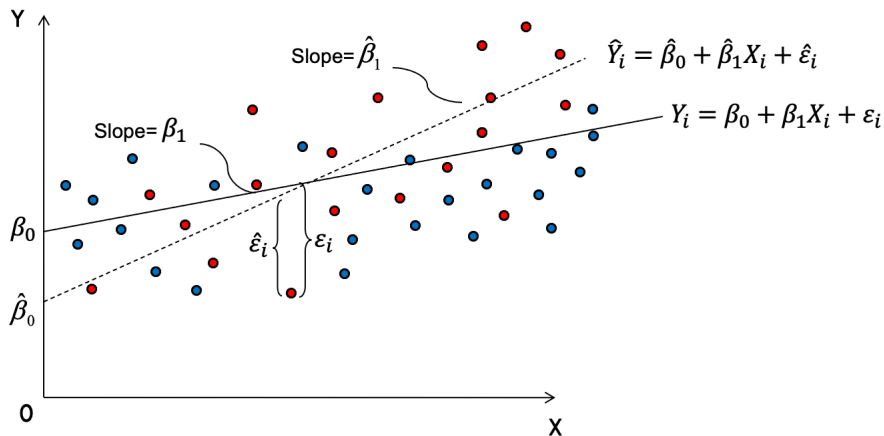
How do we interpret these coefficient estimates?

Estimating Treatment Effects - Regression Analysis

- But suppose the *population* distribution is...



Estimating Treatment Effects - Regression Analysis



Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)

Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)
- How do we know how reliable our *estimate* is likely to be?

Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)
- How do we know how reliable our *estimate* is likely to be?
 - ▶ Even if we eliminate selection bias, we're still only estimating effects for a sample and not the entire population.

Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)
- How do we know how reliable our *estimate* is likely to be?
 - ▶ Even if we eliminate selection bias, we're still only estimating effects for a sample and not the entire population.
- 1 Calculate a “standard error” for each regression coefficient (estimate of the standard deviation)

Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)
- How do we know how reliable our *estimate* is likely to be?
 - ▶ Even if we eliminate selection bias, we're still only estimating effects for a sample and not the entire population.
- 1 Calculate a “standard error” for each regression coefficient (estimate of the standard deviation)
- 2 Use the standard error to calculate a t-statistic and corresponding p -value
 - ▶ $C.I._{\hat{\beta}} = \hat{\beta} \pm t^* \times SE(\hat{\beta})$

Estimating Treatment Effects - Regression Analysis

- $\hat{\beta}_1$ is an estimate of the true β_1 , the relationship between X (the independent variable) and Y (the dependent variable)
- How do we know how reliable our *estimate* is likely to be?
 - ▶ Even if we eliminate selection bias, we're still only estimating effects for a sample and not the entire population.
- ① Calculate a “standard error” for each regression coefficient (estimate of the standard deviation)
- ② Use the standard error to calculate a t-statistic and corresponding p -value
 - ▶ $C.I._{\hat{\beta}} = \hat{\beta} \pm t^* \times SE(\hat{\beta})$
- Suppose you estimate the regression model above and find that the coefficient estimate of β_1 has a p -value of 0.115 from a two-sided t-test. What does that mean and what would you conclude about the statistical significance of the relationship between education and emergency department use?

Estimating Treatment Effects - Regression Analysis

- At this point, regression doesn't look all that different from randomization.

Estimating Treatment Effects - Regression Analysis

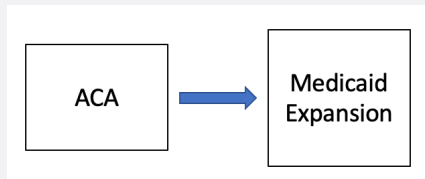
- At this point, regression doesn't look all that different from randomization.
 - ▶ In both cases we're just comparing means of treated and control groups from a sample.

Estimating Treatment Effects - Regression Analysis

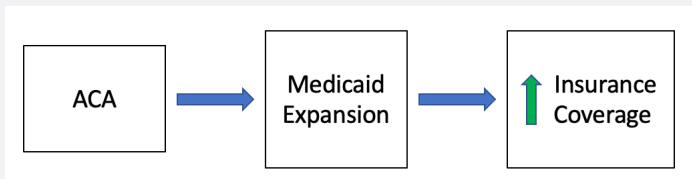
- At this point, regression doesn't look all that different from randomization.
 - ▶ In both cases we're just comparing means of treated and control groups from a sample.
- Where these methods begin to diverge is when we add additional covariates to our regression model to “control” for potential confounders.

Policy Evaluation with Regression Analysis

- Medicaid Expansion and Mortality



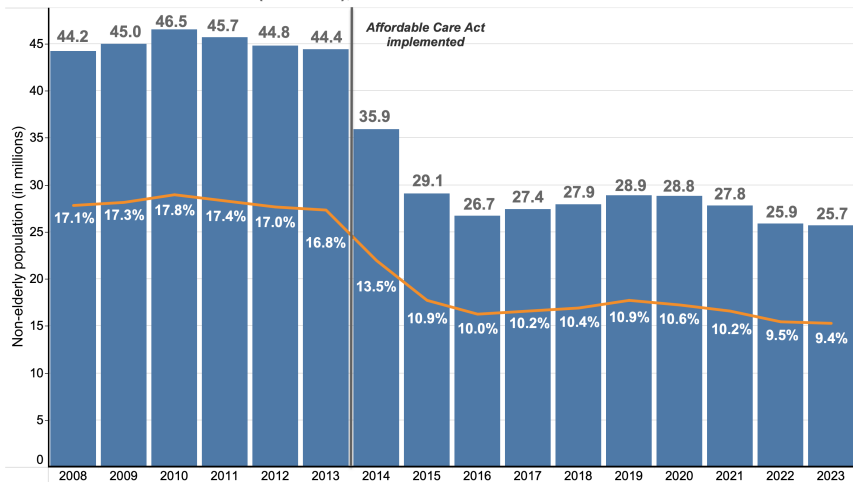
Policy Evaluation with Regression Analysis



Policy Evaluation with Regression Analysis

UNINSURED IN THE UNITED STATES

NON-ELDERLY POPULATION (AGE 0-64), NUMBER AND RATE

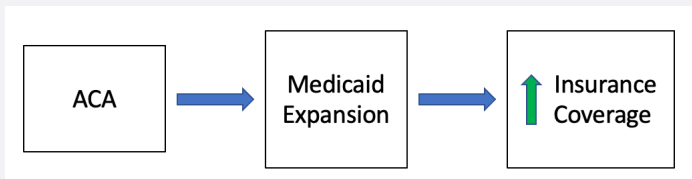


Source: Kaiser Family Foundation, analysis of 1-year ACS data

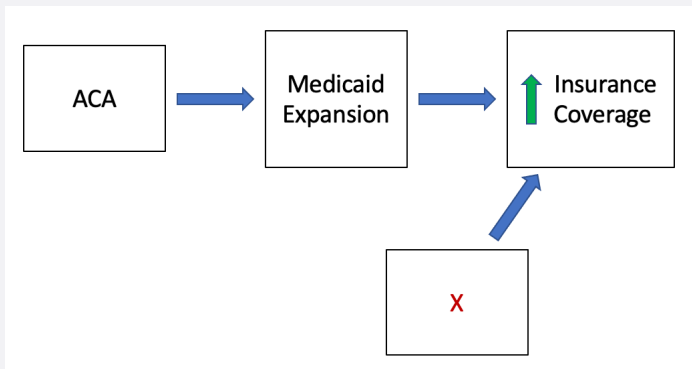
EconoFact

econofact.org

Policy Evaluation with Regression Analysis



Policy Evaluation with Regression Analysis

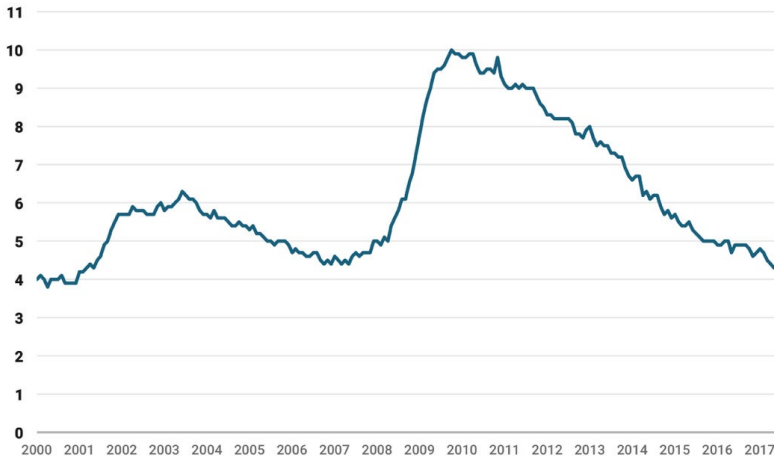


Policy Evaluation with Regression Analysis

MARKETS  CHART OF THE DAY

UNEMPLOYMENT RATE

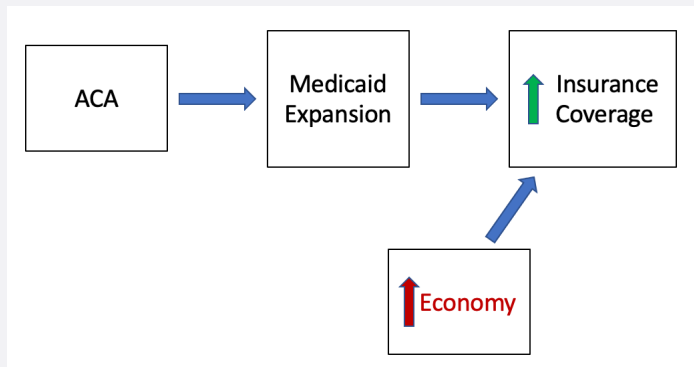
Percent of civilian labor force



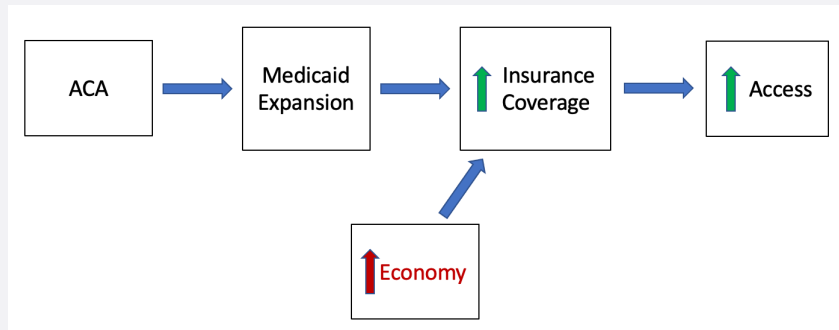
SOURCE: Bureau of Labor Statistics via FRED

BUSINESS INSIDER

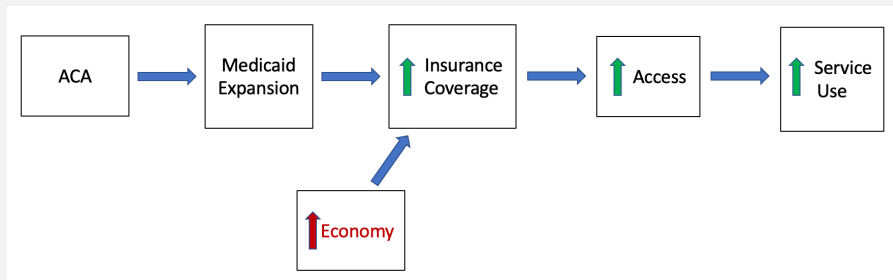
Policy Evaluation with Regression Analysis



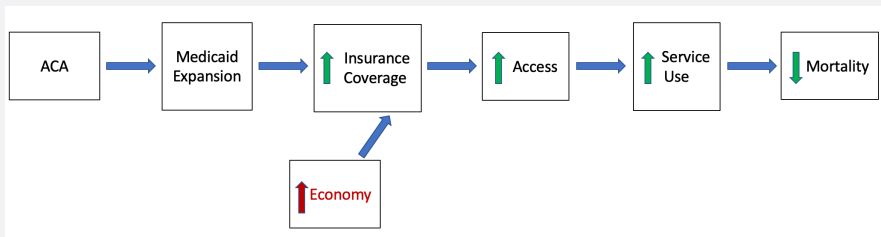
Policy Evaluation with Regression Analysis



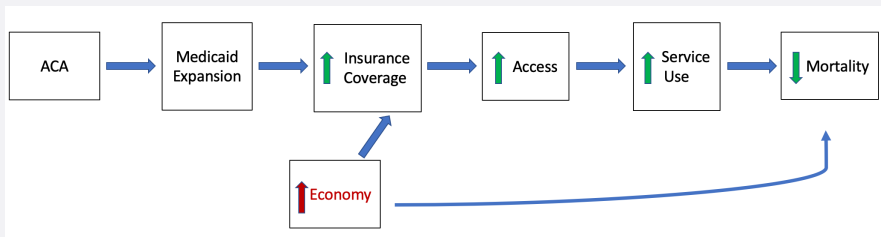
Policy Evaluation with Regression Analysis



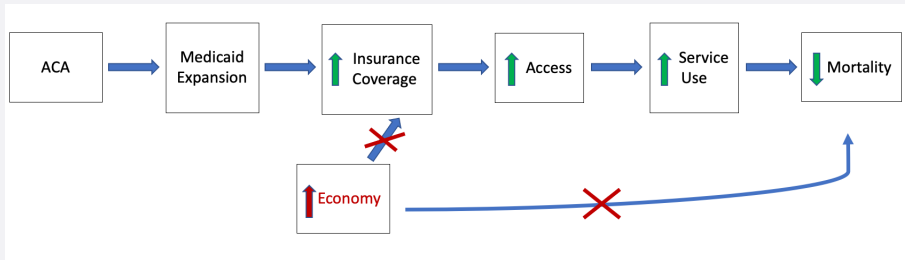
Policy Evaluation with Regression Analysis



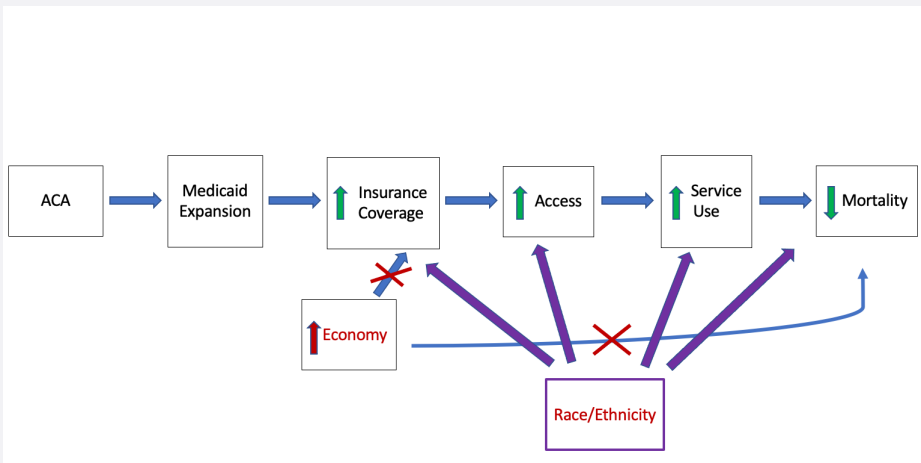
Policy Evaluation with Regression Analysis



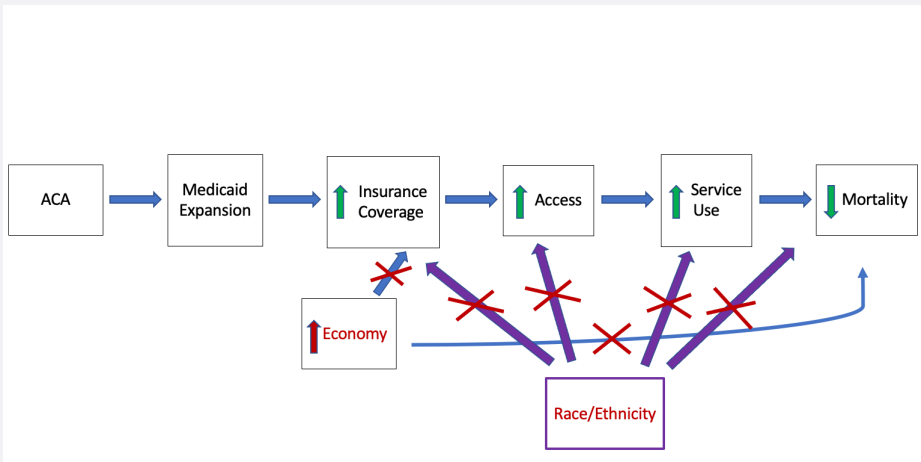
Policy Evaluation with Regression Analysis



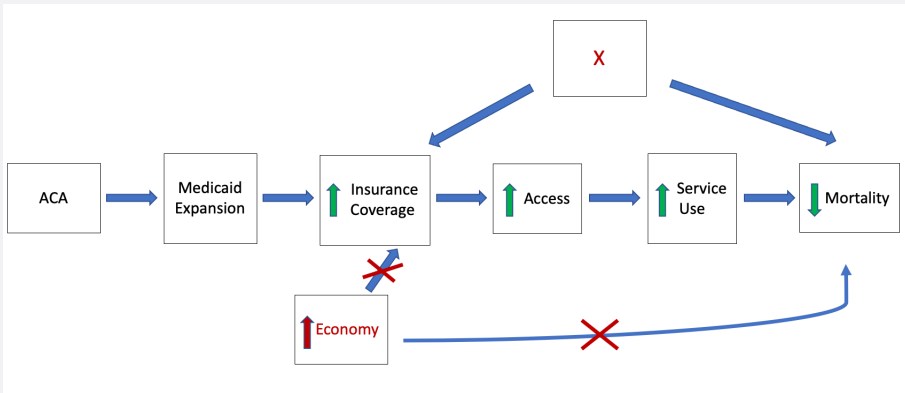
Policy Evaluation with Regression Analysis



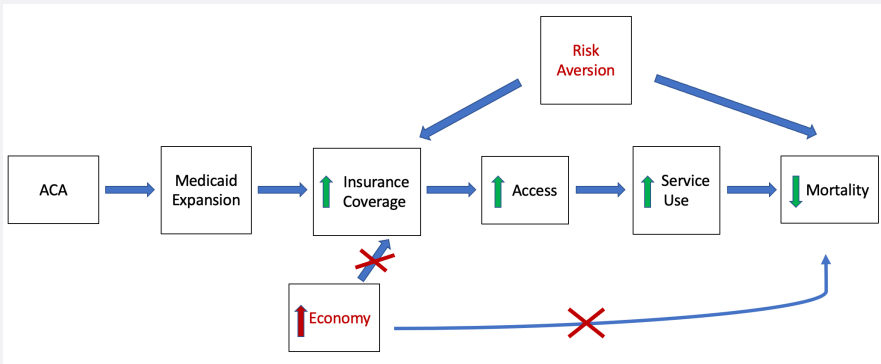
Policy Evaluation with Regression Analysis



Policy Evaluation with Regression Analysis



Policy Evaluation with Regression Analysis



Policy Evaluation with Regression Analysis

- Regression model estimating the effect of Medicaid coverage on mortality:

$$y_{it} = \beta_0 + \beta_1 D_{it} + \varepsilon_{it}$$

Policy Evaluation with Regression Analysis

- Regression model estimating the effect of Medicaid coverage on mortality:

$$y_{it} = \beta_0 + \beta_1 D_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \varepsilon_{it}$$

Policy Evaluation with Regression Analysis

- Regression model estimating the effect of Medicaid coverage on mortality:

$$y_{it} = \beta_0 + \beta_1 D_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \varepsilon_{it}$$

Policy Evaluation with Regression Analysis

- Regression model estimating the effect of Medicaid coverage on mortality:

$$y_{it} = \beta_0 + \beta_1 D_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \varepsilon_{it}$$

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \dots \beta_k X_{it} + \varepsilon_{it}$$

Policy Evaluation with Regression Analysis

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \dots \beta_k X_{it} + \varepsilon_{it}$$

- Our goal is to “control” for as many characteristics (X) as we can that could be related to insurance coverage and morality (i.e., confounders).

Policy Evaluation with Regression Analysis

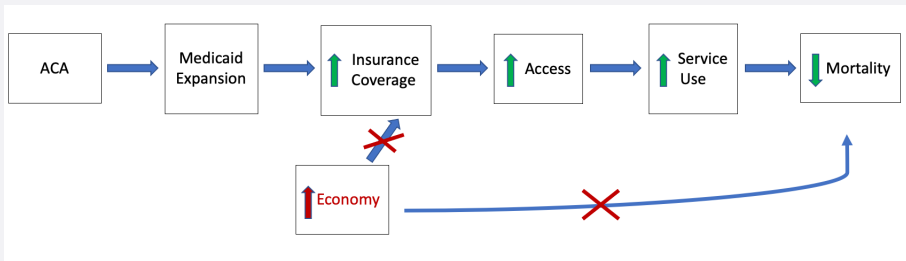
$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \dots \beta_k X_{it} + \varepsilon_{it}$$

- Our goal is to “control” for as many characteristics (X) as we can that could be related to insurance coverage and morality (i.e., confounders).
- Estimate the effect of Medicaid coverage on mortality *conditional* on education, age, etc.

Policy Evaluation with Regression Analysis

$$y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 Edu_{it} + \beta_3 Age_{it} + \dots \beta_k X_{it} + \varepsilon_{it}$$

- Our goal is to “control” for as many characteristics (X) as we can that could be related to insurance coverage and morality (i.e., confounders).
- Estimate the effect of Medicaid coverage on mortality *conditional* on education, age, etc.



Policy Evaluation with Regression Analysis

- Remember, as long as we have eliminated all relevant confounders, our regression estimate will be equal to the average treatment effect:

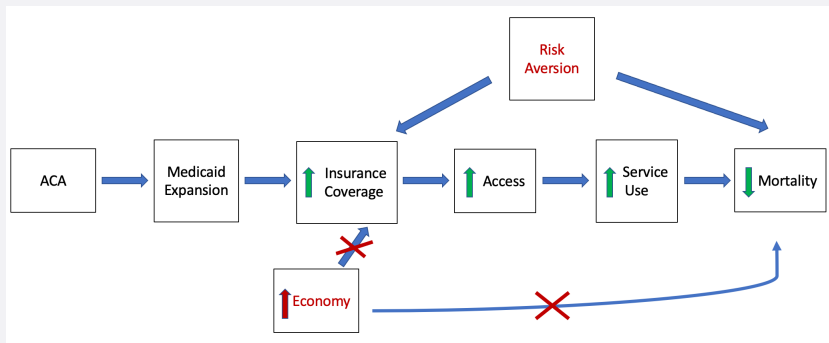
$$ATE_{est} = ATE$$

Policy Evaluation with Regression Analysis

- Remember, as long as we have eliminated all relevant confounders, our regression estimate will be equal to the average treatment effect:

$$ATE_{est} = ATE$$

- But then there's this...



Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?

Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?
- Our estimate of the effect of Medicaid coverage on mortality will be *biased*.

Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?
- Our estimate of the effect of Medicaid coverage on mortality will be *biased*.
- We think we're estimating the effect of Medicaid coverage on mortality, but we're actually estimating the combined effect of Medicaid coverage and risk aversion on mortality.

Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?
- Our estimate of the effect of Medicaid coverage on mortality will be *biased*.
- We think we're estimating the effect of Medicaid coverage on mortality, but we're actually estimating the combined effect of Medicaid coverage and risk aversion on mortality.
- Sometimes we can determine which direction that bias is likely to take.

Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?
- Our estimate of the effect of Medicaid coverage on mortality will be *biased*.
- We think we're estimating the effect of Medicaid coverage on mortality, but we're actually estimating the combined effect of Medicaid coverage and risk aversion on mortality.
- Sometimes we can determine which direction that bias is likely to take.
 - ▶ Suppose that those who are more risk averse are more likely to sign up for Medicaid coverage when eligible and are more likely to be in better health.

Policy Evaluation with Regression Analysis

- What happens if we fail to account for an observed or unobserved confounder?
- Our estimate of the effect of Medicaid coverage on mortality will be *biased*.
- We think we're estimating the effect of Medicaid coverage on mortality, but we're actually estimating the combined effect of Medicaid coverage and risk aversion on mortality.
- Sometimes we can determine which direction that bias is likely to take.
 - ▶ Suppose that those who are more risk averse are more likely to sign up for Medicaid coverage when eligible and are more likely to be in better health.
 - Then our estimate of the effect of Medicaid coverage on mortality will be biased *away from zero*.

Policy Evaluation with Regression Analysis

- Moral of the story:

Policy Evaluation with Regression Analysis

- Moral of the story:
 - ▶ Regression analysis is often used in place of randomization when randomization is infeasible.

Policy Evaluation with Regression Analysis

- Moral of the story:
 - ▶ Regression analysis is often used in place of randomization when randomization is infeasible.
 - ▶ The goal is to compare “statistically identical” people who differ only in their exposure to treatment.

Policy Evaluation with Regression Analysis

- Moral of the story:
 - ▶ Regression analysis is often used in place of randomization when randomization is infeasible.
 - ▶ The goal is to compare “statistically identical” people who differ only in their exposure to treatment.
 - ▶ Under certain conditions, regression can return causal estimates of our relationship of interest.

Policy Evaluation with Regression Analysis

- Moral of the story:

- ▶ Regression analysis is often used in place of randomization when randomization is infeasible.
- ▶ The goal is to compare “statistically identical” people who differ only in their exposure to treatment.
- ▶ Under certain conditions, regression can return causal estimates of our relationship of interest.
- ▶ But omitted variable bias is a serious threat when exposure to treatment is non-random.

Policy Evaluation with Regression Analysis

- Moral of the story:
 - ▶ Regression analysis is often used in place of randomization when randomization is infeasible.
 - ▶ The goal is to compare “statistically identical” people who differ only in their exposure to treatment.
 - ▶ Under certain conditions, regression can return causal estimates of our relationship of interest.
 - ▶ But omitted variable bias is a serious threat when exposure to treatment is non-random.
- So what do we do if randomization is infeasible and we're not confident that regression estimates will be unbiased?

Policy Evaluation with Regression Analysis

- Moral of the story:
 - ▶ Regression analysis is often used in place of randomization when randomization is infeasible.
 - ▶ The goal is to compare “statistically identical” people who differ only in their exposure to treatment.
 - ▶ Under certain conditions, regression can return causal estimates of our relationship of interest.
 - ▶ But omitted variable bias is a serious threat when exposure to treatment is non-random.
- So what do we do if randomization is infeasible and we're not confident that regression estimates will be unbiased?
 - ▶ Natural experiments and quasi-experimental research design.